# Simulation and Validation Report

16 x 8 Ternary CAM with Priority Encoder (tcam_top) — Silicluster v3, digital project, SKY130

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## 1. Circuit overview

tcam_top is a 16-entry, 8-bit-wide ternary content-addressable memory. Every stored bit is 0, 1 or X (wildcard). Given an 8-bit search word, the array compares it with all 16 stored rules in parallel. A combinational priority encoder reports the lowest-numbered matching entry (`match_addr`), whether any entry matched (`match_valid`) and whether two or more matched (`multi_match`). Rules and search words are loaded serially through a 6-bit bus, so the design needs only 14 input pins (clk, rst, op[1:0], addr[3:0], data[5:0]) and 6 output pins. It is single-clock, synchronous-reset Verilog-2001, run at 10 MHz (100 ns period).

| Pin | Dir | Width | Function |
|---|---|---|---|
| clk | in | 1 | Clock, up to 10 MHz |
| rst | in | 1 | Active-high synchronous reset; invalidates all 16 entries |
| op | in | 2 | 00 NOP, 01 LOAD_RULE, 10 LOAD_SEARCH, 11 COMMIT |
| addr | in | 4 | Entry written on COMMIT |
| data | in | 6 | Serial data chunk |
| match_addr | out | 4 | Lowest matching entry (0 when none) |
| match_valid | out | 1 | At least one entry matches |
| multi_match | out | 1 | Two or more entries match |

## 2. Circuit operation

**Cell encoding.** Each entry holds 16 bits, two per search bit {r1,r0}: 00 = X (always matches), 01 = logic 0, 10 = logic 1, 11 = invalid (never matches). An entry also has a valid flag, cleared by reset. **Writing a rule** R[15:0] to entry A takes four clock edges: three LOAD_RULE words (R[15:12], R[11:6], R[5:0]) then COMMIT with addr = A. **Searching** for S[7:0] takes two LOAD_SEARCH edges (S[7:6], then S[5:0]). The result is combinational from the stored rules and the search register, so it is valid right after the second edge and stays valid while op = NOP. While a search word is only half loaded the outputs briefly reflect a mixed word; this is visible in the waveform and is not an error.

## 3. Verification setup

The self-checking testbench `sim/tb_tcam.v` contains directed tests with explicit expected results and 400 randomized write-then-search cycles checked against an independent behavioral reference model (a separate function that computes the match vector from the rule and search values, the lowest matching index and the multi-match flag). Every randomized cycle checks match_valid, match_addr and multi_match. The testbench drives inputs on the falling clock edge and samples after the rising edge, with a 100 ns clock (10 MHz).

To reproduce (Icarus Verilog 12):

```
cd sim && bash run_sim.sh
```

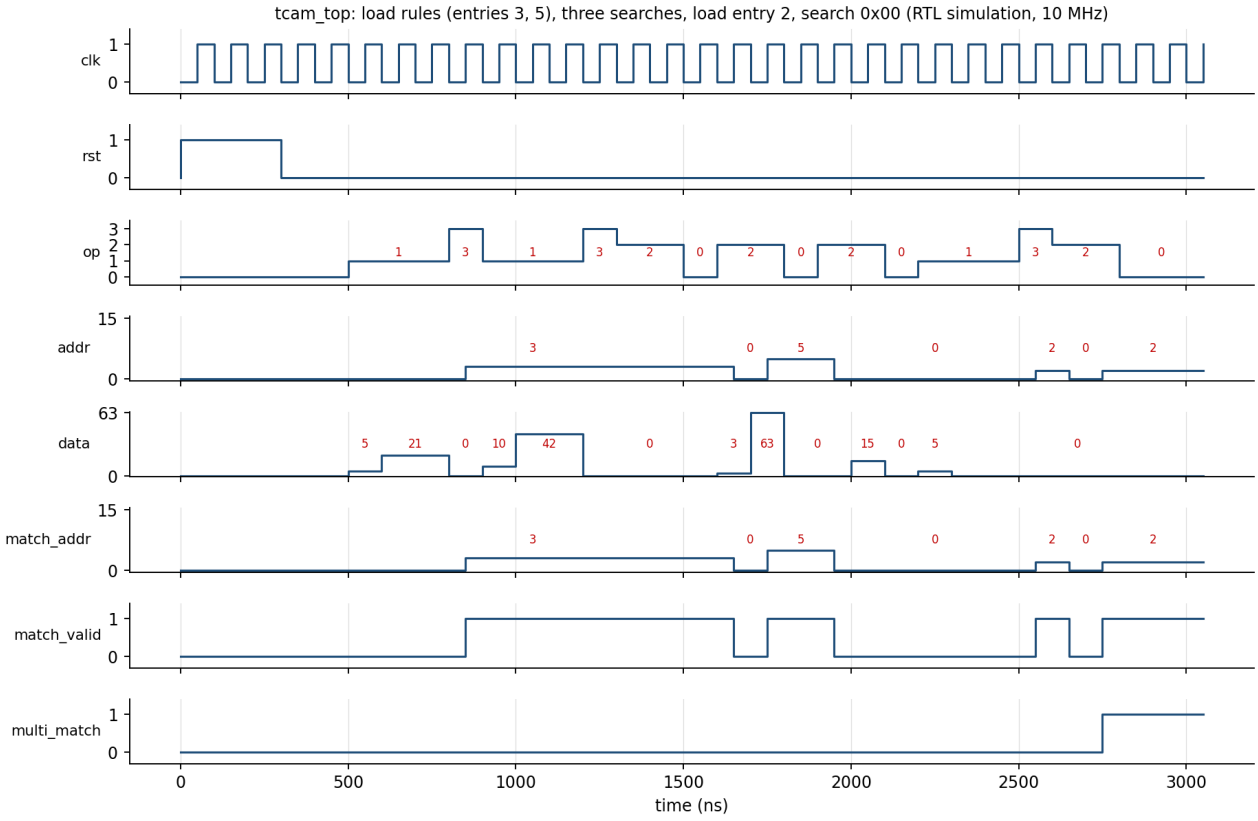
Expected last lines: `*** ALL TESTS PASSED (412 checks) ***`. The waveform in section 4 is produced by `sim/tb_wave.v` (dumps wave.vcd; sim/wave.png is the plotted result).

## 4. Results

**4.1 Functional simulation (RTL)**: 412 checks, 0 errors. The 12 directed tests and the randomized phase:

| # | Directed test (as printed by the testbench) | Result |
|---|---|---|
| 1 | T0 empty after reset | PASS |
| 2 | T1a exact 0x00 -> 3 | PASS |
| 3 | T1b exact 0xFF -> 5 | PASS |
| 4 | T1c 0x0F no match | PASS |
| 5 | T2a wildcard hit | PASS |
| 6 | T2b wildcard miss | PASS |
| 7 | T3 priority -> 2 | PASS |
| 8 | T4a invalid cell skipped | PASS |
| 9 | T4b all-X matches anything | PASS |
| 10 | T5 overwrite entry 1 (no match) | PASS |
| 11 | b overwritten entry matches 0xFF | PASS |
| 12 | T6 reset clears entries | PASS |
| 13 | 400 randomized rule/search cycles vs. reference model (outputs, priority, multi-match) | PASS |

### 4.2 **Waveform** (RTL simulation, 10 MHz)




Entries 3 (exact 0x00) and 5 (exact 0xFF) are written. Search 0x00 returns match_addr = 3; search 0xFF returns 5; search 0x0F returns no match (match_valid = 0). Entry 2 is then written with rule 00xxxxxx. Searching 0x00 now matches entries 2 and 3, so match_addr = 2 (lowest wins) and multi_match = 1. Short-lived values while a search word is half loaded are expected.

## 4.3 Physical design (SKY130)

The design was taken through the LibreLane/OpenLane flow with the files in `config.json` (PDK sky130A, 100 ns clock, absolute die area 180 x 115 um). Results come from run `RUN_2026-10-30_02-41-24` (metrics.json and the checker reports are included in evidence/):

| Item | Result |
|---|---|
| PDK / flow | sky130A, LibreLane (OpenLane classic flow) |
| Clock period | 100 ns (10 MHz) |
| Die area | 180 x 115 um (20,700 $um^2$), the area allotted by the call |
| Sequential cells | 296 (256 storage, 16 valid, 24 load/search registers) |
| Standard-cell area after routing | 12,487 $um^2$ |
| Core utilization | 65.5 % |
| DRC (Magic) | 0 errors |
| DRC (KLayout) | 0 errors |
| Routing DRC | 0 errors |
| LVS (Netgen) | 0 errors, passed |
| Antenna | 0 violating nets, passed |
| Setup / hold timing violations | 0 / 0 at the flow's timing corners |

The submission contains the final GDS (layout/tcam_top.gds), the Magic layout (layout/tcam_top.mag), and the Magic DRC, KLayout DRC and Netgen LVS reports plus metrics.json in evidence/.

## 5. Scope and limitations

Functional results are from zero-delay RTL simulation. No gate-level or post-layout timing simulation is included; timing is covered by the static timing checks of the physical-design flow. Pin assignments follow the call: up to 14 inputs, 14 outputs, one clock, one reset, 10 MHz maximum.